

Cognitive control in older Minnan-Mandarin and Hakka-Mandarin bidialectal adults: Advantages in Stroop-type tasks[☆]

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Abstract

The effect of bidialectalism on older bidialectal adults' performance on cognitive tasks has received little attention in aging studies. The current research aimed to investigate older monolingual and bidialectal adults' performance on nonverbal and verbal cognitive tasks, including Simon, Stroop, flanker, and spatial n-back tasks. Bidialectalism and task effects were examined via a comparison of the monolinguals' and bidialectals' response latencies. Two experiments were conducted, which consisted of four nonverbal tasks (Experiment 1) and four verbal tasks (Experiment 2), respectively. The participants were 20 older Mandarin monolingual adults and 20 older Minnan-Mandarin/Hakka-Mandarin bidialectal adults from Taiwan. The results indicated that the bidialectals performed better than the monolinguals on the nonverbal and verbal Stroop color-word tasks and the verbal Stroop day-night task. The nonverbal tasks revealed no significant interaction of task and language group, while the verbal tasks showed the opposite results. These findings suggest that the advantages in cognitive control extended beyond bilingualism to bidialectalism. Moreover, the bidialectal advantage was found in some tasks which involved more attentional and inhibitory control and had a high or intermediate level of task difficulty, and was absent in other tasks.

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Keywords: Effect of bilingualism; Effect of bidialectalism; Task effect; Task difficulty level; Cognitive control; Inhibitory control

Introduction

Over the past few decades, bilingual effects have been extensively explored in psycholinguistics and the topic has become increasingly important, particularly in light of present-day demographic changes (e.g., Blom et al., 2014; Blumenfeld and Marian, 2014; Cox et al., 2016; Desideri and Bonifacci, 2018; Goral et al., 2015; Kousaie and Phillips, 2017; Kroll and Bialystok, 2013; Mertz, 2017). As the world's population is aging quickly, bilingualism may furnish shield against age-related decline in executive functioning (also known as cognitive control abilities) (Anderson et al., 2018; Bak et al., 2014; Craik and Bialystok, 2005; Diamond, 2013; Fernandes et al., 2007; Kavé et al., 2008). Numerous studies have

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reported that the advantage of bilingualism has often been found in nonverbal tasks (e.g., Simon and flanker tasks), while a bilingual disadvantage has frequently been identified in verbal tasks (e.g., picture-naming and read-aloud tasks) (Bialystok, 1999; Bialystok, 2009; Bialystok et al., 2008; Blom et al., 2014; Blumenfeld and Marian, 2014; Braver, 2012; Gollan et al., 2007; Hsu, 2017; Oller and Eilers, 2002, to name a few).¹ In contrast to the study of effects of bilingualism on cognition, however, how bidialectalism modulates cognitive control capacities has been a neglected area of investigation. Without understanding how cognitive processes work in bidialectals, doubts about whether bidialectalism is an influential factor in research outcomes cannot be excluded. The objective of this study was to determine whether bidialectal experience benefits older bidialectal adults' cognitive control, and if so, to find out which tasks identified bidialectal advantages and how their effects shaped bidialectals' performance.

Bilingualism has been found to advantageously influence executive functioning, for example, attention, inhibition, monitoring, and working memory (Bialystok, 1999; Bialystok et al., 2003; Blom et al., 2014; Blumenfeld and Marian, 2014; Braver, 2012; Colzato et al., 2008). For instance, Costa and colleagues (2009) carried out a flanker task and analyzed the behavioral performance of 122 young adults, and the results showed that bilinguals yielded shorter response times than their monolingual counterparts did in the high-monitoring condition,² suggesting a bilingual advantage in cognitive control, particularly the monitoring process. This positive effect is generally designated a bilingual advantage in the previous studies (Bialystok et al., 2003; Costa et al., 2009; Hilchey and Klein, 2011). Bilingual advantages result from bilinguals' constantly directing their attention to the target language, maintaining and monitoring the language goal, and suppressing interference from the irrelevant language because when speaking either of them, bilingual individuals use language control mechanisms to continuously manage the two activated languages in order to produce the target language fluently and successfully, (Costa et al., 1999; Hoshino and Thierry, 2011; Kroll et al., 2008). Bilingual speakers are able to switch from one language to the other smoothly in bilingual conversations. Bilinguals' increased use of language control mechanisms is considered an advantage in cognitive control, such as monitoring, attentional control, and inhibitory control (Bialystok, 2011; Bialystok et al., 2009; Foy and Mann, 2013; Hsu, 2014, 2017; Kroll and Bialystok, 2013).

The claimed cognitive advantages possessed by the bilinguals have significant implications to the healthy aging process for elder populations. A good many studies have manifested that bilingual advantages can persist across a bilingual's lifespan, and that bilingualism can slow age-related decline in cognitive control and even delay the onset age of dementia (Anderson et al., 2018; Bak et al., 2014; Bialystok et al., 2004; Bialystok et al., 2007; Craik and Bialystok, 2005; Fernandes et al., 2007; Kavé et al., 2008; Valenzuela and Sachdev, 2006). In past studies, advantages in cognitive control have been identified in bilinguals' performing executive functioning tasks, while these advantages were not found among monolinguals, and they were more pronounced in older adults than in younger adults, suggesting that bilingualism can provide protection against age-associated decline in executive functioning (Bialystok et al., 2008). For example, Bialystok and colleagues (2004) carried out three studies and reported that bilingualism attenuated normal cognitive decline in older adults. Bialystok et al. (2004) compared the performance of middle-aged monolingual and bilingual adults and older adults using a Simon task to test executive functioning and found that both bilingual age groups responded faster than their monolingual counterparts when the task imposed greater demands on cognitive control, suggesting that bilingualism was associated with smaller Simon-effect costs. Interestingly, the benefit of bilingualism was greater for older adults in all cases in Bialystok et al. (2004), who claimed that bilingual individuals executed controlled processing more effectively and that bilingualism compensated for cognitive control losses in certain executive processes due to aging. In a later study, Bialystok et al. (2007) investigated whether lifelong bilingualism could retain cognitive functioning and provide a buffer against the onset of symptoms of dementia in old age. Bialystok et al. (2007: 462) recruited 184 patients diagnosed with dementia in a memory clinic, about half of whom were bilinguals, and from the sample, they found that bilingual patients displayed "a delay of 4.1 years in the onset of symptoms of dementia" as compared with their monolingual counterparts.

Although bilingual advantages were commonly found in nonverbal tasks, bilingual disadvantages were found in verbal tasks. For instance, Gollan and colleagues (2005) examined the effect of bilingualism by conducting two picture-naming experiments in which English-speaking monolinguals and English-Spanish bilinguals participated and found that compared with the monolinguals, the bilinguals named pictures in their dominant language – either in English or in both languages – more slowly and produced more errors. These bilingual disadvantages in verbal tasks were explained as a result of heavier linguistic processing load in the verbal tasks for the bilinguals, rather than as a challenge to the validity of the cognitive advantage possessed by the bilinguals. Bilinguals performed poorly, as compared with monolinguals, in verbal tasks, because the bilinguals' vocabulary size was approximately twice as large as that of the monolinguals and their two languages were used unequally in daily conversations, leading to the fact that the bilinguals experienced greater difficulty in accessing and retrieving words in their weaker language (Gollan et al., 2007; Hsu, 2017; Oller and Eilers, 2002;

¹ Regarding the difference between verbal and nonverbal tests, a verbal test is a language production task in which words are produced, and a nonverbal test is a perceptual task in which stimuli, such as words or pictures, are perceived rather than uttered.

² Bilinguals and monolinguals had similar performance in the low-monitoring condition (Costa et al., 2009).

Ransdell and Fischler, 1989). With such an uneven frequency of using two languages, the link between lexical and semantic representations in either language was weaker for the bilinguals than for the monolinguals (Gollan and Acenas, 2004; Gollan and Silverberg, 2001). Given that the lexical access and retrieval is a decisive factor resulting in bilingual disadvantages and it is excluded from experiments, does the disadvantage still exist in the verbal cognitive tasks? Stated differently, can bidialectal advantages be recognized in the verbal tasks in which the lexical factor is ruled out (if bilingual advantages can be extended to bidialectalism)?

A good deal of past literature has reported that bilingual experience advances bilingual speakers' cognitive control, which in turn mitigates age-related cognitive decline. Whether such an enhanced cognitive control can be identified in bidialectal speakers is an important question nowadays, particularly for the present situation where the world population is aging rapidly and the world's bidialectal population accounts for a large proportion. However, this issue is largely unexplored as no cognitive study on bidialectalism is conducted.

Generally speaking, bilingualism refers to the ability to speak two different languages. In contrast, bidialectalism refers to the ability to speak two different dialects. The mutual intelligibility and political and social conventions are two criteria most used to distinguish a language from a dialect in the studies of bilingualism and bidialectalism (Chambers and Trudgill, 1998; Cheshire, 2007; Edwards, 2007; Francis, 2016; Hazen, 2001; Lee-James and Washington, 2018; Lyons, 1981; Milroy, 1987). Based on the former, languages are not mutually intelligible and dialects are mutually intelligible. Dialects commonly refer to closely related varieties of a particular language and are used by a particular group of the language's speakers. According to the latter, two languages are politically or institutionally recognized which may be mutually intelligible to a great degree (e.g., Danish and Norwegian) and the dialects of a language may be not mutually intelligible (e.g., the dialects of Chinese). The code of political and social conventions is often used to discriminate bilingualism from bidialectalism in the research on cognitive control (Bialystok et al., 2008; Kirk et al., 2014; Kousaie and Phillips, 2012; Salvatierra and Rosselli, 2011). Mandarin, Minnan, and Hakka are viewed as the closely related dialectal varieties of the Chinese language even though they are mutually unintelligible. In a certain sense, the so-called dialects of the Chinese language are like separate languages (e.g., English versus Spanish). However, these variants share a large proportion of words which are lexical cognates, but these cognates have considerably different sounds.³ For example, the character “北 (north)” is pronounced as “pei” in Mandarin, as “pak” in Minnan, and as “pet” in Hakka. Interestingly, these dialects also share the same writing system which is based on Mandarin grammar (so that bidialectal speakers hardly are aware of the differences in linguistic components between the two dialects they have acquired). On this view, the variants of the Chinese language are not completely like separate languages (e.g., English versus Mandarin). Therefore, it is impossible for us to readily apply the previous finding that the cognitive advantages were found in bilinguals to the bidialectal case. With this reasoning, it aroused our special interest in exploring whether the cognitive advantage in bilingualism could be extended to bidialectalism. The effect of long-term managing two dialects has received little attention in cognitive studies because dialects are in general treated as tightly related variants of a language and share more linguistic properties (e.g., lexical items, phonology, and syntax) relative to separate languages;⁴ in other words, the linguistic difference between dialects is considered much less than that between separate languages. This raises the following questions: what effect (positive or negative) on cognitive control is evoked by the continuous management of two dialects and to what extent does it impact bidialectal speakers' cognitive control?

Consequently, one limitation resulting from prior studies is that the elderly bilinguals recruited in previous research on cognitive control, except for Kirk and colleagues' (2014) research, were individuals who spoke two languages rather than two dialects,⁵ for example, Mandarin-English (Bialystok et al., 2008), Tamil-English (Bialystok et al., 2004), English-French (Kousaie and Phillips, 2012), and Spanish-English (Salvatierra and Rosselli, 2011). Thus far few studies have tapped into the effect of bidialectalism on executive functioning tasks (e.g., Simon tasks) and the present study attempts to address this research void by studying if there is any cognitive advantage on the part of the Minnan-Mandarin and Hakka-Mandarin bidialectals as compared with the Mandarin-speaking monolinguals.⁶ According to the social and political

³ This may be a reason why the variants of the Chinese language are not mutually intelligible.

⁴ As asserted by Oswald et al. (2018), bidialectalism is generally considered “an extreme form of language similarity” and dialects are closely associated with their originating language and share more linguistic properties.

⁵ Precisely, the languages that the participants spoke had different writing systems and were unequivocally considered separate languages.

⁶ In Taiwan, a majority of the people mainly communicate in either Mandarin or Minnan or both, while a minority of Taiwanese speak Hakka, or other Chinese dialects, or Austronesian in public. Mandarin is the official language of Taiwan and is the dominant language in education and public domains. All of the individuals who participated in the current study were educated at school with the same writing system and script; specifically, when these participants were educated at school in Taiwan, they learned the writing system which was based mainly on Mandarin grammar rather than Minnan or Hakka grammar, and they learned Chinese characters which correspond to the sound system of Mandarin Chinese, but not Minnan or Hakka. Thus it was difficult for them to be aware of the differences in linguistic properties (e.g., lexemes, syntactic structure, and phonotactics) between Minnan and Mandarin as well as between Hakka and Mandarin, as related by our participants in the interviews.

criteria, Minnan-Mandarin and Hakka-Mandarin individuals are classified as a bidialectal group in this study and the terms “bidialectals” and “bidialectalism” are used when referring to the participants in the study and the resulting bidialectal effects, respectively.

Although bidialectals constitute the majority of the population in some areas and countries (e.g., Taiwan), the studies of the relationship between bidialectalism and cognitive control abilities have been scant, while bilingualism has been focused on and extensively investigated. Given that bilinguals are found to have cognitive advantages in cognitive executive functioning, it is highly likely that such cognitive advantages can also be found among the bidialectals with the associated benefits for healthy and successful aging. Consequently, understanding how bidialectalism affects older bidialectal adults' cognitive capacities becomes a critical issue for the current situation in which the population is aging worldwide. Every country in the world is experiencing a dramatic growth in the number and proportion of senior citizens in their population; for example, according to the estimated statistical data for 2010 to 2060, the population of Taiwan will largely be an aging society.⁷ More importantly, over recent decades, a growing number of individuals have suffered from neurocognitive disorders such as dementia. These demographic changes in the elderly population and the consequent cognitive control abilities of older adults are becoming issues of great importance in delaying age-related cognitive decline and its effect on healthcare systems. Increasing life expectancy without healthy and adequate cognitive abilities will impose a considerable negative effect on individuals and higher costs on the society. Therefore, whether bidialectal experience benefits older adults in retaining cognitive control is an important issue.

Apart from a dearth of investigation on bidialectalism in past literature, another limitation is the task types utilized in a study. Although conflicting findings were found regarding bilingual advantages in nonverbal versus verbal tasks, where bilingual advantages were found in nonverbal tasks, and bilingual disadvantages were found in verbal tasks, very few studies used both verbal tasks and nonverbal tasks simultaneously to examine the bilingual effects. Furthermore, most of the previous research on bilingualism conducted either only one single or two executive control tasks (e.g., Simon and Stroop tasks) to examine bilingual (dis)advantages in cognitive control (see, among many others, Bialystok et al., 2004; Bialystok et al., 2006; Coderre and van Heuven, 2014; Kirk et al., 2014; Kousaie and Phillips, 2012; Poarch and van Hell, 2012; Rosselli et al., 2000).⁸ For example, Gollan and colleagues (2005) utilized a verbal picture-naming task to explore bilingual effects; in Ansaldo et al.'s (2015) study, only a Simon task was conducted; and Blumenfeld and Marian's (2014) study carried out both Simon and Stroop tasks. Consequently, the full profile of bilingual effects on certain cognitive control processes (e.g., inhibitory control) has yet to be thoroughly revealed. Using a single task or a type of task to measure cognitive control cannot capture the complete profile of bidialectal cognitive (dis)advantages since executive control consists of different components, such as monitoring, inhibitory control, and attentional control, rather than only a single component. In this study, we will examine the performance differences between monolinguals and bidialectal in verbal and nonverbal tasks and the effect of task difficulty levels on speakers' performance. It will be difficult to design the types of tasks by utilizing only one task and manipulate the task difficulty levels by using only one or two tasks. Therefore, we will adopt multiple tasks in the current research in order to answer these questions.

Thus little research has directly explored the subtle processing differences in cognitive control between monolinguals and bidialectals by adopting a sequence of different but related tasks. Task processing demand is one of the determinants in the emergence of bilingual advantages, as claimed in Dong and Liu's (2016) study, and therefore, using a set of tasks allows us to prevent the floor and ceiling effects from occurring as much as possible (Costa et al., 2008; Van den Noort et al., 2019).⁹ The current study adopted both verbal and nonverbal tasks and a series of different levels of task difficulty to examine bidialectal effects and aimed to explore whether the advantages in cognitive control extended beyond bilingualism to bidialectalism.

In order to examine bidialectal effects, the present study adopted and designed four nonverbal tasks (i.e., Simon, Stroop Color and Word, flanker, and spatial n-back) and four verbal tasks (i.e., Number Stroop, Stroop Color and Word, Stroop day-night, and Stroop picture-naming) on the basis of Van den Noort and colleagues' (2019) findings and those in earlier studies reviewed in this study (e.g., Bialystok et al., 2004; Bialystok et al., 2008; Blumenfeld and Marian, 2014; Costa et al., 2006; Costa et al., 2009; Costa et al., 2008). Van den Noort and colleagues (2019) reviewed and analyzed

⁷ The data from the Council for Economic Planning and Development of Executive Yuan (CEPT) reported that the number of senior citizens is on the rise: the overall population of 65-year-old people and above will increase from 10.7% in 2010 to 41.6% in 2060; among the overall population, 80-year-old people will increase from 24.4% in 2010 to 44% in 2060. This is a clear result of an aging society. The report is available at the following websites: http://cdnet.stpi.narl.org.tw/techroom/policy/2012/policy_12_017.htm; and http://www.twfact.org/index.php/2018_%E5%8F%B0%E7%81%A3%E4%BA%BA%E5%8F%A3%E9%9B%B6%E6%88%90%E9%95%B7.

⁸ As reviewed by Van den Noort and colleagues (2019), although past studies that examined discrepancies between monolinguals and bilinguals conducted a variety of executive control tasks, most of them adopted only one or two tasks to compare bilinguals' and monolinguals' performance, resulting in the investigation of only certain components of cognitive control.

⁹ As Hilchey and Klein (2011) pointed out, numerous studies on bilingual advantages in inhibitory control used divergent designs and methodologies, which might have led to conflicting results.

46 original studies and found that bilingual advantages in cognitive control were more evident in Stroop and flanker tasks, while more divergent and less convergent evidence supporting bilingual benefits was observed in Simon and other types of cognitive control tasks. [Van den Noort and colleagues \(2019: 19–20\)](#) further claimed that Simon tasks were among “the easier cognitive control tasks,” while Stroop tasks were among “the more difficult cognitive control tasks.” Both the easier tasks and the more difficult tasks have been adopted in the current study to investigate how the levels of task difficulty influence monolinguals’ and bidialectals’ performance in terms of cognitive control.

To date, prior research has scarcely been focused on the manipulation of the difficulty levels among cognitive tasks to explore executive functioning in bilinguals and bidialectals, such that how the relative degrees of difficulty between tasks impact bilinguals’ and bidialectals’ performance remains greatly unreported. Consequently, the present study, to the best of our knowledge, is a first attempt to design a series of related nonverbal and verb tasks ranked from the lowest to the highest levels of difficulty in terms of the processing demands for cognitive control for older adults. The difficulty levels of the tasks were identified and manipulated in the current study based on the findings of earlier studies, [Kornblum et al.’s \(1999\)](#) dimensional overlap model,¹⁰ and two pilot studies we conducted before this study. The difficulty level of a task is defined as how much cognitive control is involved in the task and how much effort is required to respond to or name the test items, as measured by the response time. The tasks utilized in the present study involved subtly different attentional, inhibitory, and working memory components; they differed in cognitive demands and processing loads. In what follows, these tasks will be presented and discussed, particularly focusing on the relative levels of difficulty in which the tasks differed from each other.

First, the classic Simon task was used in this study, which measures stimulus-response inconsistency using two different forms as stimuli. It involves two stimulus dimensions that have no overlap: color is used as the target dimension, whereas the distractor dimension is the location on the screen, meaning that divergent forms are utilized. Conflict in a Simon task is created by the inconsistency or overlap between the irrelevant stimulus (e.g., left-side location) and correct response (e.g., right key), which results in a stimulus-response conflict ([Kornblum, 1994; Kornblum et al., 1999](#)). To effectively perform a Simon task, participants must remember the stimulus-response rule and overcome the conflict through the inhibition of the predominant response, which is induced by the spatial position of the stimulus. Accordingly, attentional control and inhibitory control are important components in this task.¹¹

Different from the Simon task, the classic Stroop task, namely the Stroop Color and Word Test, measures stimulus-stimulus incompatibility, which involves two stimulus dimensions and employs the same form—words—for both dimensions. The target dimension is the color-word’s ink color and the distractor dimension is the color-word’s name, both of which completely overlap and result in a stimulus-stimulus conflict ([Kornblum, 1994; Kornblum et al., 1999](#)). Stroop interference originates from competition and conflict between word-reading and color-naming, and the prepotent word-reading must be restrained to efficiently and correctly name the target stimulus, that is, the incongruent color of the ink ([Bialystok et al., 2008; MacLeod, 1991](#)). Greater Stroop interference discloses weaker interference inhibition. In this sense, inhibitory control and attentional control have their own role in the Stroop task, whereas more resources tend to be prioritized for inhibition and others are left for the operation of attentional control. In view of the target-distractor separability and the perceptual stimulus-stimulus conflict (the target and non-target stimuli merged into the same stimulus), the classic Stroop task is more difficult than the classic Simon task.

In a similar vein, a flanker task is categorized as a stimulus-stimulus conflict test as well because it uses the same form of stimulus—arrows—for its target and distractor dimensions (e.g., $\leftarrow \rightarrow \leftarrow \rightarrow$), both of which resort to the same response type, either right or left ([Blumenfeld and Marian, 2014](#)). However, different from the classic Stroop task, in which the distractor and target dimensions converge on a single word, the target arrow and interfering flankers are not conflated on the same stimulus; instead, the target and the to-be-ignored flankers in the flanker task are separate. To complete this task successfully, participants need to resolve the conflict between the target and the distractor stimuli by ignoring the irrelevant flankers and directing their attention to the target stimulus. Hence, attentional control and inhibitory control play a crucial role in this task; to be precise, the previous reports suggest that the flanker task involves attentional control much more than inhibitory control. In terms of attentional focus, the flanker task is easier than the classic Simon and Stroop tasks.

From the perspective of attentional focus, the flanker task is easier to perform when compared with the classic Simon and Stroop tasks because its target and distractor occupy distinct spatial locations and participants can reduce distractor interference by skillfully adjusting and narrowing their attentional focus on the target location.¹² However, in the classic Stroop tasks, the target and distractor are merged into the same stimulus, so that it is relatively difficult to rule out

¹⁰ This model accounts for a variety of cognitive control processes and mechanisms.

¹¹ As pointed out by [Bialystok and colleagues \(2005: 2\)](#), “[a] nonverbal task that can be used to examine the extent to which bilinguals and monolinguals differ in inhibitory control is the Simon task, an experimental paradigm based on stimulus-response compatibility.”

¹² In the flanker task, participants merely need to focus mainly on the central target arrow and ignore the distracting flanked arrows.

interference from the distractor no matter whether attentional focus is expanded or narrowed (Chen, 2003; Eriksen and James, 1986; LaBerge et al., 1991; Yantis and Johnston, 1990). In the classic Simon task, participants need to unceasingly move their attentional focus to the right or left side of the screen where the current stimulus is being presented; at the same time, they have to inhibit the interference from the spatial position of the stimulus. In this view, the classic Simon and Stroop tasks are more difficult tests than the flanker task.

The Number Stroop task integrates the characteristics of the flanker task and the classic Stroop Color and Word Test. It contains two dimensions, the digits themselves and the number of digits (e.g., 111), causing a partial overlap of the target and distractor stimuli. The distractor in this task is digit-reading, which is more predominant than the target, the number of digits. To name the number of digits correctly and efficiently, participants must direct their attention to the number of digits by inhibiting interference from the digits themselves, suppressing irrelevant responses, and ruling out conflicting information. Thus, this task is also a measure of cognitive control, particularly attentional control and inhibitory control. In terms of the dimensional overlapping degree (see Table 1 below), the Number Stroop task is more difficult than the classic Simon and flanker tasks but easier than the classic Stroop task in either a nonverbal or a verbal manner.

Stroop picture-naming tasks in general involve not only working memory but also cognitive control. For example, the Stroop day-night task, a kind of picture-naming task, relies heavily on inhibitory control to suppress a habitual or predominant response (Martin-Rhee and Bialystok, 2008). The present research adopted the Stroop day-night and Stroop picture-naming tasks from Gerstadt and colleagues' (1994) study and Costa and colleagues' (2006) study, respectively. In the Stroop day-night task, when shown a picture framed by a blue circle in congruent trials, the participants must say aloud the name of the picture; in incongruent trials, when shown a picture framed by a red circle, they must say aloud the opposite name of the picture. To complete the task successfully, participants must not only remember the naming rules but also inhibit habitual or prepotent responses in the incongruent trials. For the Stroop picture-naming task utilized in the current study, when the picture is presented in blue, the participants have to say aloud its Mandarin name; if the picture is red, they do not have to name it but instead say aloud *meiyou* “沒有 (no)” in Mandarin. Similar to the rules for the Stroop day-night task, to perform the task smoothly, the participants had to remember the rules and restrain their impulse to say the names of the pictures under the condition of saying *meiyou* “沒有 (no)” in Mandarin. According to the requirements of

Table 1

A summary of the dimensions and relative difficulty levels of the tasks.

	Task	Major executive functions measured	Target vs. distractor dimensions	Dimensional overlapping degree	Difficulty level ^a
Nonverbal	Flanker	Attentional Control > Inhibitory Control	Same form: Centered arrow vs. flanked arrows	Perceptually separate	Low
	Simon Color-Space	Inhibitory Control $\cong$ Attentional Control	Different forms: Color of object vs. position of color object	Perceptually separate	Low+
	Spatial N-back	Working Memory > Inhibitory Control (small part)	Position of target circle in a triangle	Perceptually separate (asynchronous appearance of target & distractor)	Intermediate
	Nonverbal Stroop Color-Word	Inhibitory Control > Attentional Control	Color-word's ink color vs. color-word's name	Totally overlapping	High
Verbal	Number Stroop	Attentional Control $\cong$ Inhibitory Control	The number of the digits vs. digit	Partially overlapping	Low
	Verbal Stroop Color-Word	Inhibitory Control > Attentional Control	Color-word's ink color vs. color-word's name	Totally overlapping	Intermediate
	Stroop Day-Night	Working Memory $\cong$ Inhibitory Control	Opposite name of picture vs. name of picture	Only picture (word retrieval required)	High
	Stroop Picture-Naming	Working Memory $\cong$ Inhibitory Control	Saying “No” vs. name of picture	Only picture (word retrieval required)	High

Note: The symbol “>” = greater than (in amount); “ $\cong$ ” = approximately equal to (in amount); “+” = higher.

^a The difficulty levels were compared among the four nonverbal tasks and among the four verbal tasks, respectively.

the two tasks, it was difficult for us to unequivocally differentiate their comparative difficulty levels. Consequently, based on previous studies (Gerstadt et al., 1994; Costa et al., 2006) and the major traits of the two tasks, the two tasks were considered similar in terms of degree of difficulty.

Compared with the Stroop Color and Word task, the Stroop picture-naming and day-night tasks are more difficult to perform in accordance with the WEAVER++ model (Levitt et al., 1999; Roelofs, 2003, 2007).¹³ Conversely, the Number Stroop task is easier to complete because its stimulus display consists of a target and a distractor that partially overlap, and thus interference from distracting information is reduced by expanding the spatial extent of attentional focus (Chen, 2003; Eriksen and James, 1986; LaBerge et al., 1991; Yantis and Johnston, 1990). Accordingly, the processing demands for these variants of the Stroop tasks are different. Among them, the Stroop picture-naming and Stroop day-night tasks are more difficult than the Stroop Color and Word task and the Number Stroop task.

Quite different from all of the aforementioned tasks, the n-back task is a continuous-recognition task frequently utilized to measure working memory (WM)¹⁴ capacity in cognitive neuroscience (Braver et al., 1997; Perlstein et al., 2001; Teubner-Rhodes et al., 2016). Although the task is primarily a working memory task, it involves conflict processing and needs specific cognitive functions such as recall, attention, and “inhibition” (Diamond, 2013; Rac-Lubashevsky and Kessler, 2016).¹⁵ In this task, participants must compare the upcoming stimulus with the stored stimulus in WM to decide whether or not the stimuli are the same to make a correct response. To complete this task successfully, the participants need to encode perceived items, temporarily store the stimulus sequence (e.g., letters) in WM, and continuously update incoming stimuli; also, they must simultaneously inhibit irrelevant information and discard the current non-target items from WM (Kane and Engle, 2003). Grounded on the previous research, the present study created a spatial 1-back task. In this task, a trial consists of two white circles and one blue circle on a white background on a computer screen. The three circles are arrayed in a triangle, and the blue circle is placed in one of the three angles of the triangle randomly. The participants have to determine whether or not the blue circle is located in the same place as that in the previous trial. Accordingly, to perform this task correctly, the participants must not only remember the position of the blue circle in the triangle but also “inhibit” interference from its location in the triangle, particularly when it is positioned in the left or right side which corresponds to the left or right response key, respectively (see Fig. 2(c)). At the same time, they must “inhibit” previous task set, namely to “suppress” the recently stored item in trial *n*. In this view, the spatial 1-back task requires inhibitory control. However, the n-back task does not rely on inhibitory control as heavily as other executive functioning. The present study adopted the n-back task with a low level of inhibition to manipulate the amount of inhibition to a certain degree and further examine how the demand for inhibitory control affects the participants’ performance in cognitive control tasks.

Taken together, regarding the difficulty levels of the four nonverbal tasks used in the present study, the Simon, Stroop Color and Word, and flanker tasks are mainly measures of cognitive control, particularly attentional and inhibitory control, while the n-back task relies more heavily on working memory but partly on inhibitory control. The nonverbal tasks employed in the current study involve divergent cognitive processes and loads, leading to different degrees of difficulty. For example, the flanker task contains two perceptually separate dimensions, thereby requiring less capacity of cognitive control, whereas the Stroop Color and Word task requires higher capacity as it bears a high perceptual load (i.e., two perceptually overlapping dimensions) that consumes more perceptual capacity; therefore, the difficulty level of the flanker task is lower than that of the Stroop Color and Word task.

With respect to the difficulty levels of the four verbal Stroop-type tasks adopted in this study, they are ranked from low to high as Number Stroop (low), Stroop Color and Word (intermediate), Stroop day-night (high), and Stroop picture-naming (high). These verbal tasks involve different cognitive processes and loads, thereby bringing about different degrees of difficulty. For instance, the Number Stroop task involves two partially overlapping dimensions and requires less capacity of inhibitory control relative to the Stroop Color and Word task; accordingly, the difficulty level of the former is lower than that of the latter.

The different cognitive dimensions involved in all the tasks used in this study, and the relative difficulty levels of these tasks are summarized in Table 1 below.

¹³ Grounded on the WEAVER++ model, a lexical processing model (Levitt et al., 1999; Roelofs, 2003, 2007), as soon as pictures are perceived, concepts are immediately activated, and following the activation of concepts, word forms are activated, whereas when words are perceived, word forms can be directly retrieved and activated without activating concepts. This means that words can be spoken by simply seeing the word forms (e.g., the visual word “pen”) and articulating them (e.g., ⟨pen⟩ and /p/, /e/, /n/) without mentioning their concepts. However, when perceiving pictures, participants have to retrieve and activate their concepts first and then the word forms, which may result in their making more of an effort in or spending more time performing the picture-naming tasks compared with the word-naming and ink-color-naming tasks.

¹⁴ Redick and Lindsey (2013:1110) stated that “WM is a multifaceted system that relies on multiple processes (encoding, maintenance, recall, recognition, familiarity, updating, temporal ordering, binding, attention, and inhibition).”

¹⁵ Miller and colleagues (2009:711) clearly pointed out that “the n-back is not a pure measure of working memory, but may be able to detect subtle differences in cognitive functioning between PD patients and controls.”

The earlier studies on cognitive control reviewed above have demonstrated the presence of bilingual advantages on executive functioning in different tasks. Based on the findings of the past literature, we can only speculate that the cognitive advantage emerges among bilinguals whose languages are not mutually intelligible. However, little is known about whether the cognitive advantage found in bilinguals can be observed in elderly bidialectals. Therefore, this study investigates whether there is any cognitive advantage among the elder bidialectal adults. Furthermore, previous research on bilingual effects mostly adopted one or two cognitive control tasks in a single study and recognized bilingual advantages in various cognitive tasks performed by different groups of participants rather than a single group, such that heterogeneous factor cannot be excluded. To address this issue, this study provides a within-subject design across different types of cognitive task types (verbal tasks versus nonverbal tasks), and across different task difficulty levels within each task type (verbal versus nonverbal). Accordingly, four specific research questions will be raised, as follows:

- (a) Will bidialectal advantages emerge when the older bidialectal adults perform cognitive nonverbal tasks, resulting in a better performance compared with their monolingual counterparts?
- (b) Do the difficulty levels of the nonverbal tests influence the monolinguals' and bidialectals' performance in the tasks?
- (c) Will bidialectal advantages emerge when the older bidialectal adults perform cognitive verbal tasks, resulting in a better performance compared with their monolingual counterparts?
- (d) Do the difficulty levels of the verbal tests influence the monolinguals' and bidialectals' performance in the tasks?

To answer these four questions, the present study adopts four nonverbal and four verbal cognitive tasks and compares the processing differences in response times between older monolingual and bidialectal adults. Moreover, four hypotheses are put forth in accordance with previous literature.

Hypothesis 1 (:).

Older bidialectal adults will perform better with faster responses than their monolingual peers in the nonverbal cognitive tasks which involve more inhibitory control and attentional control.

Hypothesis 2 (:).

The more demand for inhibitory control and attentional control a nonverbal task requires (i.e., the higher the difficulty level), the more pronounced the bidialectal advantage will be.

Hypothesis 3 (:).

Older bidialectal adults will perform worse with slower response time than their monolingual peers on verbal tasks.

Hypothesis 4 (:).

The more demand for inhibitory control and attentional control a verbal task requires (i.e., the higher the difficulty level), the more pronounced the bidialectal advantage will be.

Method

Experiment 1: nonverbal tasks

The nonverbal tasks in Experiment 1 test how monolingual and bidialectal adults perform when a task does not depend heavily on lexical retrieval but perception and recognition; conversely, the verbal tasks in Experiment 2 test how monolingual and bidialectal adults perform when a task mainly involves lexical access and retrieval in the speech production. By comparing older monolingual and bidialectal adults' performance on nonverbal tasks, this study explored how bidialectal experience shaped the processing of different cognitive tasks and modulated attentional and inhibitory control mechanisms to resolve conflicts arising in different tasks with diverse difficulty levels. Because the degree of task difficulty is a determinant factor in the emergence of bilingual advantages (Costa et al., 2009; Dong and Li, 2015), based on the dimensional overlap model (Kornblum et al., 1999) and Van den Noort and colleagues' (2019) study, the difficulty levels of the tasks in the current study were manipulated by adopting Simon, flanker, spatial n-back, and Stroop tasks. The difficulty level of a task was defined as how much cognitive control was needed in the task and how much effort was required to respond to stimuli. The difficulty levels of the tasks were ranked from low to high in the following order: flanker task (low), Simon Color and Space task (low+), spatial 1-back task (intermediate), and nonverbal Stroop Color and Word task (high). It is assumed that bidialectals will outperform monolinguals in nonverbal tasks and when the tasks require greater inhibitory control and attentional control, the bidialectal advantage is expected to be more likely to emerge.

Participants

Forty right-handed participants took part in the nonverbal experiment. The participants grew up and lived in the southern part of Taiwan. All of them were literate and educated in Mandarin as it is the official language in Taiwan. Their ages ranged from 61 to the early 70s. They were recruited by responding to an advertisement on Facebook, Line, the Bulletin Board System, and the like that sought participants for a study on cognitive control. They were classified into two groups based on their language background: Mandarin monolinguals (13 males/7 females, mean age = 67.75 years, SD = 4.34 years, range = 61.0–73.0 years) and unbalanced bidialectals¹⁶ (both D1 Minnan-D2 Mandarin and D1 Hakka-D2 Mandarin, 20 participants).¹⁷ The bidialectal group consisted of 10 Minnan-Mandarin adults (7 males/3 females, mean age = 68.00 years, SD = 4.71 years, range = 61.0–74.0 years) and 10 Hakka-Mandarin adults (6 males/4 females, mean age = 68.70 years, SD = 4.50 years, range = 62.0–75.0 years), and their less dominant dialect¹⁸ was D2 Mandarin.¹⁹ All of the participants were required to have normal or corrected-to-normal vision. None of them had speech or hearing disorders and neurological or psychiatric disorders, and all of them had normal scores (above 24 points) on the Mini-Mental Status Exam (MMSE), which was used to exclude dementia. An independent samples *t*-test was administered and showed no significant differences between the groups in terms of age and MMSE scores.

Moreover, all of the participants were interviewed and invited to complete a questionnaire composed of demographic information, language history, and language proficiency self-assessment to ensure that they were eligible. The study adopted a 5-point Likert scale—1 (poor), 2 (fair), 3 (good), 4 (very good), and 5 (excellent)—to allow the bidialectal individuals to self-evaluate their own listening, speaking, and reading proficiency in Mandarin.²⁰ Only bidialectals who scored more than three points on all three linguistic abilities in their self evaluations were included in this experiment.²¹ These individuals had mastered their two dialects at different levels of proficiency and thereby were categorized as “unbalanced” D1 Minnan-D2 Mandarin and D1 Hakka-D2 Mandarin bidialectals; none of the participants scored five points in all three listening, speaking, and reading abilities in Mandarin, which would have placed them in the “balanced” category. Concerning the use of D2 Mandarin, these participants were seldom exposed to bidialectal conversations in which they needed to switch back and forth between their two dialects; instead, they mostly used D1 at home and spoke D2 outside the home (e.g., in the workplace or in some public areas). The participants’ demographic and neurological characteristics are displayed in Table 2 below (see Appendix A for more details on the language background of the Minnan-Mandarin and Hakka-Mandarin bidialectal participants).

Apparatus

This experiment used E-prime (version 2.0), which was administered on an ASUS PC with a 14-inch computer monitor, to program the tasks. Through this software, practice and experimental stimuli were presented. The participants’ response times were recorded via the response box. On the response box, the right and left buttons were labeled “X” and “O,” respectively.

Materials and design

Experiment 1 was composed of four nonverbal tasks that were designed on the basis of different demands for processing the task. Each task contained one block, and therefore, four blocks were constructed for this experiment. In the

¹⁶ Although they were viewed as “unbalanced” bidialectals, they had a native speaker’s competence in both dialects.

¹⁷ Corresponding to the term “bidialect,” D1 and D2 were used in this study, referring to the first dialect (more dominant) the participants acquired and the second dialect (less dominant), respectively.

¹⁸ Although the bidialectal group showed a native speaker’s competence in both dialects, one of their two dialects is more dominant than the other.

¹⁹ In the current study, the performance of the two groups, monolinguals and bidialectals, was analyzed. The latter was composed of both Minnan-Mandarin and Hakka-Mandarin participants; that is, Minnan-Mandarin and Hakka-Mandarin participants did not constitute two separate analysis groups in this study.

²⁰ The participants’ language proficiency levels were classified in accordance with their self-assessment in the questionnaire. The questions included in the questionnaire were created based on the listening, speaking, and reading assessments provided by the Interagency Language Roundtable (ILR) (available at <https://www.govtllr.org/>). For each assessment, the questions were divided into five sections (Level 1 to Level 5), corresponding to the five points of the 5-point Likert scale – 1 (poor), 2 (fair), 3 (good), 4 (very good), and 5 (excellent). For example, in the speaking assessment, the participants answered Level 1 questions such as “I can ask and answer gender, nationality, date and place of birth, marital status, and occupation,” “I can order a simple meal,” and “I can tell someone how to reach the place (he or she asked) that is near here.” After completing the self-assessment, the participants were interviewed, which examined whether the participants’ language proficiency levels coincided with their self-assessment.

²¹ There were four participants in the D1 Minnan-D2 Mandarin group and six in the D1 Hakka-D2 Mandarin group whose listening and speaking scores were 4 or above; the other participants’ scores were 3.

Table 2

Demographic and neurological characteristics of the elderly participants.

	Monolingual (N = 20)	Bidialectal(N = 20)	Independent samples <i>t</i> -test		
			<i>df</i>	<i>t</i>	<i>p</i>
Male-to-female ratio	M13:F7	M13:F7			
Age range (in years)	61.00–73.00	61.00–75.00			
Mean age (in years)	67.75 (4.34)	68.35 (4.50)	38	-.429	.751
D2 AOA (in years)		11.50 (2.12)			
Daily use of D2 (percent)		40.5%			
D1 proficiency	5.00 (.00)	5.00 (.00)			
D2 proficiency		4.15 (.37)			
	Listening	3.50 (.51)			
	Speaking	3.25 (.44)			
	Reading				
MMSE	26.80 (.83)	26.40 (.99)	38	1.378	.218

Note: N = number of participants; standard deviations in parentheses; D2 AOA = onset of D2 acquisition; MMSE = Mini-Mental State Examination; D2 proficiency = participants' self-assessed language proficiency in three language skills (listening, speaking, and reading) in total points.

experiment, the order of the presentation of the nonverbal tasks was counterbalanced across participants. The four tasks for cognitive control were ranked from low to high based on task demands in the following rank order: flanker task (low), Simon Color and Space task (low+), spatial 1-back task (intermediate), and nonverbal Stroop Color and Word task (high).

- (a) *Flanker task*. This task was adapted from Eriksen and Eriksen (1974). The stimuli were arrows—a central arrow flanked on each side by two arrows. This task comprised 40 trials, half of which were congruent and half incongruent, and they were randomly presented. A trial included five arrows. The central arrow appeared along with four flanker arrows pointing in the same (congruent trials →→→→→) or different (incongruent trials ←←→→→) directions. The participants were instructed to determine whether the central arrow pointed to the right or to the left. If the arrow pointed to the right, the participants had to press the O button on the response box; if it pointed to the left, they had to press the X button.
- (b) *Simon color and space task*. In this task, the conditions manipulated were color and space. The task was composed of 40 trials: half were congruent and half were incongruent, which were presented on the screen in random order. When a blue circle appeared, the participants were asked to press the O button, and when a red circle appeared, they were asked to press the X button. In the congruent trials, the position of the blue circle corresponded to that of the O button and that of the red circle to the X button. In the incongruent trials, the positions of the circles and the response buttons were opposite; specifically, the position of the blue circle corresponded to that of the X button and the red circle to that of the O button.
- (c) *Spatial 1-back task*. This 1-back task was designed in light of previous research (Dong and Liu, 2016; Teubner-Rhodes et al., 2016). This task consisted of 30 trials and the stimuli were blue and white circles. A trial contained two white circles and one blue circle on a white background on the screen. The three circles were arrayed as a triangle and the distance between them was equal. In each trial, the blue circle appeared in one of the three angles of the triangle randomly. The participants were required to identify whether or not the position of the blue circle in the current trial appeared in the same place as that in the previous trial. If the two trials were the same, they had to press the O button, otherwise they had to press the X button.
- (d) *Nonverbal Stroop color and word task*. The stimuli were color-word names (such as black, blue, and red), which were printed either in the same ink color as the names of the color-words or in ink colors different from the names of the color-words. This task contained 40 trials, half of which were congruent and half were incongruent. In each trial, two words were presented. The ink color of the word on the left side of the screen was black, while the ink color of the word on the right side was black, purple, blue, red, or green. Regarding the words on the right side, in some trials, the name of the color-word and its ink color were congruent, for example, the word “紫” denoting “purple” was displayed in purple ink; in other trials, the name of the color-word and its ink color were incongruent, for example, the word “綠 (green)” was displayed in red ink. The participants were instructed to decide whether the name of the color-word on the left side matched the ink color of the color-word on the right side, by pressing the O button for congruent pairs and the X button for incongruent pairs.

The schema for the stimuli and the presentation of each task is illustrated in Fig. 1 below.

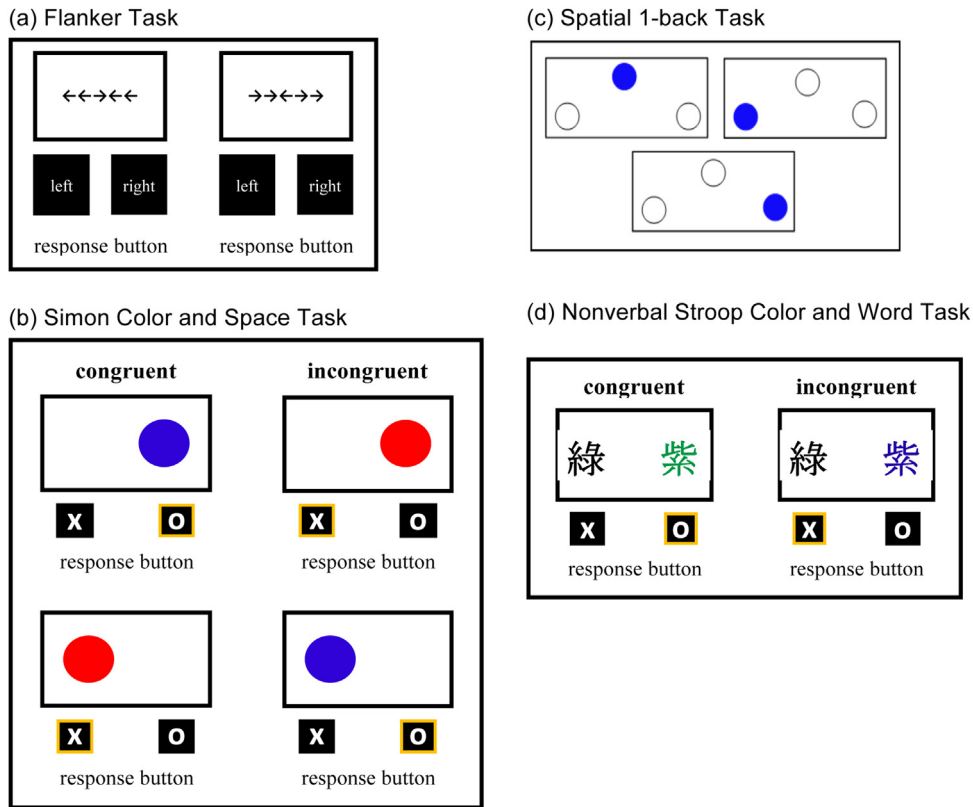


Fig. 1. Schematics of the four nonverbal tasks in Experiment 1 (Note: illustration (b) shows the three positions of the blue circles; in illustration (e), the words “綠” and “紫” mean “green” and “purple,” respectively.) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Procedure

Before the experiment, all of the participants were given an informed consent, which they signed upon agreeing to take part in the experiment, and they were told that their rights were protected under the human research ethics of National Cheng Kung University in Taiwan. They were tested individually in a soundproof booth and compensated with \$25 for participating in the experiment. The testing of each participant lasted approximately one and a half hours and included four five-minute breaks between the tasks.

At the beginning of each of the tasks, the participants were given the following instructions read by the experimenter and displayed on the computer screen:

In this experiment, you will see []²² on the computer screen. After seeing the stimuli, decide on your answer and press the response button without hesitation and without worrying about producing any errors.

In each block, a fixation cross “+” will appear on the center of the screen and remain for two seconds. Your attention needs to be focused on the fixation point.

Respond to each stimulus as quickly and as accurately as you can. You may also take as much time as you like before pressing the button. After you press the button, the current trial will disappear and the next trial will appear. There will be 10 consecutive practice trials to accustom you to this procedure.

After the instructions were given, the participants practiced 10 or more trials to familiarize themselves with the task's procedure. The practice trials were followed by the experimental trials.

In the four tasks, all of the stimuli were presented one at a time on a 14-inch screen and remained on the screen until the participant responded. Each task constituted a block. For each block, the trial presentation was as follows: (1) a fixation

²² In brackets, words such as “circles” in the Simon task and “arrows” in the flanker task were inserted.

point (“+”) was shown on the center of the screen and remained for 2000 milliseconds (ms); (2) 10 practice trials appeared in a row, each practice item disappeared after the participant's response, and the next trial started; (3) after the 10 practice trials, the experimental stimuli appeared one by one, and each stimulus remained on the screen until the participant pressed the response button and the next stimulus appeared; and (4) after the presentation of the last stimulus in each block, an asterisk “*” was presented for 1000 ms, informing the participant of the end of the block. Fig. 2 below shows the schematic illustration of the presentation of the stimuli in each task:

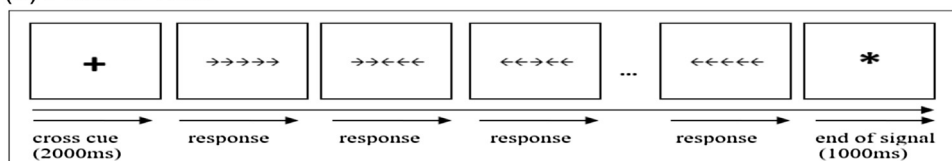
Measurement of response latencies

The participants' response times (RTs) were obtained by the response box connected to the computer. For each stimulus, the RTs were calculated from the onset of its trial to the participant's pressing of the response button. The time unit was milliseconds (ms). Overall RTs were calculated as the total RTs of all valid trials divided by the aggregate number of valid trials. Only times above 215 ms were treated as valid trials based on Indefrey and Levelt's (2004) speech production and perception model and Sage's (1995) reaction time theory.

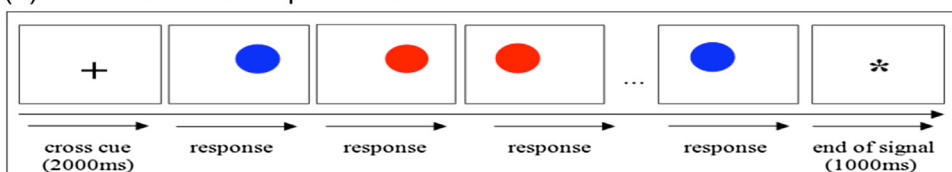
Results of Experiment 1

To examine whether there was a statistically significant difference in RTs between the two language groups across all four nonverbal tasks, a two-way 2 (language groups: monolingual vs. bidialectal, between subjects) $\times$ 4 (different tasks, within subjects repeated measures) mixed ANOVA was administered using IBM SPSS Version 20.0. The four nonverbal tasks included flanker, Simon Color and Space, spatial 1-back, and nonverbal Stroop Color and Word (henceforth, Flanker, Simon CS, 1-back, and Stroop CW(nonv)). The error data were further inspected and it was found that the monolinguals and bidialectals did not differ significantly in error rates ($p > .05$), thereby excluding the speed-accuracy tradeoff as a factor in performance.

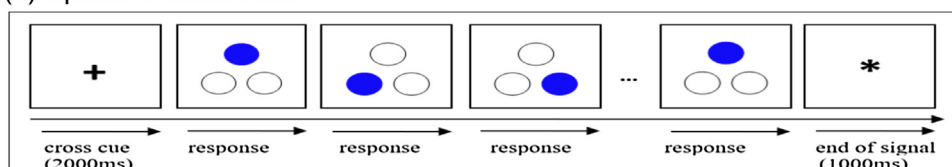
(a) Flanker Task



(b) Simon Color and Space Task



(c) Spatial 1-back Task



(d) Nonverbal Stroop Color and Word Task

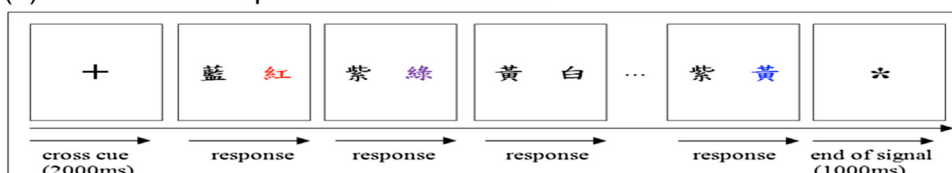


Fig. 2. Schematics of the presentation of the stimuli in the four nonverbal tasks in Experiment 1.

Table 3
The two-way ANOVA table for Experiment 1.

Source	df	MS	F	p	η^2
Language Group $\times$ Tasks	3	12727.66	1.911	.132	.048*
Language Group (independent factor)	1	44455.56	1.495	.229	.038*
Tasks (dependent factor)	3	2043962.91	306.89	.000*	.890

Note: * $p < .05$.

A two-way ANOVA for language groups (2) and tasks (4) showed no significant interactions between the two factors, $F(3, 114) = 1.911$, $p = .132$, indicating that the performance differences between the monolinguals and bidialectals were not affected by the different difficulty levels of diverse tasks. The main effect of task was significant, $F(3, 114) = 306.89$, $p < .001$, $\eta^2 = .890$, suggesting that participants' performances were influenced by the task difficulty levels. The main effect of language groups was not significant, $F(1, 114) = 1.495$, $p = .229$, $\eta^2 = .038$, suggesting there were no performance differences between the two language groups when the performances of the four tasks were collapsed. The statistical results are reported in Table 3 below.

The Bonferroni post-hoc tests on the task effect indicated significant differences in RTs among the four nonverbal tasks ($p < .001$), except for between Flanker and Simon CS ($p = .541$). The RTs for Stroop CW(nonv) ($M = 1074.35$ ms) were significantly longer than those for the other three tasks ($p < .001$ for all pairwise comparisons). The RTs for 1-back ($M = 696.50$ ms) were significantly shorter than those for Stroop CW(nonv) ($p < .001$), but longer than those for Flanker ($p < .001$) and Simons CS ($p < .001$). The difference of RTs between Flanker and Simons CS was not significant. These results indicated that Stroop CW(nonv) had the highest difficulty level, 1-back had the intermediate difficulty level, and Flanker and Simons CS had the low difficulty level. These results are summarized in Tables 4 and 5.

The main effect of language group was not significant, $F(1, 114) = 1.495$, $p = .229$. Post-hoc pairwise comparisons showed no significant difference between the two language groups for Simon, Flanker, and 1-back tasks. However, there was a marginally significant difference between the two language groups in the Stroop CW(nonv) task (monolingual M : 1115.25 > bidialectal M : 1033.45; $p = 0.047$) (see Table 6). The bidialectals responded more quickly than the monolinguals did, indicating a bidialectal advantage on Stroop CW(nonv), which had the highest difficulty level among the four nonverbal tasks. Although this post-hoc pairwise comparison result cannot be used to draw valid conclusions regarding the hypotheses of this study, it provides a good indicator relevant to the potential interaction between language background and task difficulty levels for future studies, which will be explored in Discussion.

Table 4
Mean response times of the monolingual and bidialectal participants in the four nonverbal tasks.

Difficulty level		Monolingual ($N = 20$)		Bidialectal ($N = 20$)		All Participants ($N = 40$)	
		Mean (ms)	SD	Mean (ms)	SD	Mean (ms)	SD
T1-Flanker	Low	587.45	126.54	582.95	110.00	585.20	18.75
T2-Simon CS	Low+	634.00	100.41	595.55	106.56	614.78	16.37
T3-1-back	Intermediate	700.80	99.83	692.20	90.53	696.50	15.07
T4-Stroop CW(nonv)	High	1115.25	137.44	1033.45	113.26	1074.35	19.91

Note: N = number of participants; ms = millisecond, the unit for response latencies.

Table 5
The pairwise comparisons for the task effects in Experiment 1.

Tasks		MD	SD	p
Flanker	Simon CS	-29.575	17.007	.541
	1-back	-111.300*	16.089	.000*
	Stroop CW(nonv)	-489.150*	20.172	.000*
Simon CS	1-back	-81.725*	14.803	.000*
	Stroop CW(nonv)	-459.575*	20.158	.000*
1-back	Stroop CW(nonv)	-377.850*	20.436	.000*

Note: MD = Mean Deviation; SD = Standard Deviation; * $p < .05$.

Table 6

The pairwise comparisons for the language effects for each task in Experiment 1.

Task	Language Group		MD	SD	<i>p</i>
Flanker	Monolingual	Bidialectal	4.500	37.493	.905
Simon CS	Monolingual	Bidialectal	38.450	32.739	.248
1-back	Monolingual	Bidialectal	8.600	30.134	.777
Stroop CW(nonv)	Monolingual	Bidialectal	81.800*	39.824	.047*

Note: MD = Mean Deviation; SD = Standard Deviation; **p* < .05.

Discussion of Experiment 1

In Experiment 1, four nonverbal cognitive measures were employed to investigate whether there were significant differences in people's response times in the cognitive control related tasks between the older monolingual and bidialectal adults and whether the task difficulty levels modulated their response times. Mixed results were obtained, partially confirming earlier studies. The results showed that there was no significant interaction between task difficulty levels and language groups. There was a main effect of task difficulty levels, showing that the four nonverbal tasks indeed represented different task difficulty levels to the participants. Although there was no significant main effect of language group, in the post hoc pairwise comparisons, the bidialectal group showed significantly faster response times than the monolingual group for the Stroop CW(nonv) task, while there were no significant language group differences for the other three easier tasks (the Simon, Flanker, and 1-back tasks).

The finding that no language group difference was found in the Simon, Flanker, and 1-back tasks is consistent with the previous studies that reported the absence of bilingual advantages on Simon tasks in children (Morton and Harper, 2007), in young adults (Naeem et al., 2018), and in older adults (Kirk et al., 2014), as well as on Simon and flanker tasks in young adults (Paap and Greenberg, 2013) and on Simon, flanker, and n-back tasks in middle-aged adults (Van der Linden et al., 2018). This finding, together with the cited earlier findings, does not lend support to the bilingual advantage account. However, a significant difference was identified in Stroop CW(nonv) in the current study between the monolinguals and the bidialectals. Since Stroop CW(nonv) was more difficult than the other three tasks, the findings in Experiment 1 suggest that the bidialectal advantages only showed up in difficult tasks, but not in easy tasks. This result is consistent with those found in previous research which found bilingual advantages in inhibition control (Blumenfeld and Marian, 2014; Colzato et al., 2008; Goral et al., 2015; Poarch and van Hell, 2012).

In the present study, bidialectal advantages did not exist across different levels of task difficulty. Bidialectal advantages emerged only in Stroop CW(nonv), the most difficult task in Experiment 1. In order to perform such a difficult task successfully, participants have to use attentional resources skillfully and effectively because the attentional resources are finite. The finding of bidialectal advantage in the nonverbal Stroop Color and Word test suggests that bidialectals had better capacity to utilize attentional resources more efficiently than monolinguals.

This finding from the nonverbal Stroop task intrigued us to further explore whether the bidialectal advantage identified in the nonverbal Stroop CW task can be observed in "verbal" Stroop tasks. In the studies of bilingual speakers, bilingual "disadvantages" are generally recognized in the verbal tasks. The bilingual disadvantages in verbal tasks were hypothesized to primarily result from greater difficulty in accessing and retrieving lexical items (Gollan and Acenas, 2004; Gollan et al., 2007; Michael and Gollan, 2005; Oller and Eilers, 2002). Given that the interference from lexical access has been reduced from verbal tasks to a certain degree,²³ it is expected that the negative effect of bilingualism on lexical process might be mitigated and its positive effect on cognitive control might be more likely to emerge. The Stroop Color and Word task is a case in point. Precisely, the Stroop CW consists of a limited set of words, from which participants benefit to make less of an effort to access and retrieve lexical items, as compared with such tasks whose stimuli are not "words" as semantic verbal fluency tasks (e.g., Gollan et al., 2002).²⁴ Furthermore, as compared with the tasks such as semantic verbal fluency tasks involving a wide range of related words, a Stroop picture-naming task which contains a limited number of pictures allows speakers to make less effort to access and retrieve lexical items when participants familiarize themselves with target words for the pictures. In this view, the negative influence of the lexical access and

²³ For example, the test lexical items are provided in advance and in so doing, the lexical accessibility limits are more or less decreased to a certain degree.

²⁴ "Semantic verbal fluency task is a task to produce as many words belonging to a specific category as possible within a given time" (Oh et al., 2019:48). In Gollan and colleagues' (2002:276) study, participants had to produce examples on three category types, namely "12 semantic, 10 letter, and 2 proper name fluency," as many as possible.

retrieval, which causes the bilingual disadvantage, might be diminished in such verbal Stroop tasks as Stroop Color and Word and Stroop picture-naming. If that is the case, then the bidialectal advantage observed in the nonverbal Stroop may also show up in such verbal tasks which involve relatively limited lexical items. This observation has given rise to the question of whether the pattern of the bidialectal advantage obtained in the nonverbal tasks (Experiment 1) can be observed in verbal tasks.²⁵ To answer this question, a second experiment was conducted. In Experiment 2, four versions of “verbal” Stroop tasks were constructed by manipulating the different processing demands for cognitive control, in order to examine how the levels of the task difficulty affected the monolinguals’ and bidialectals’ performances in the verbal tasks.

Experiment 2: verbal Stroop tasks

This experiment consisted of four versions of verbal Stroop tasks in which the participants needed to inhibit habitual and prepotent responses. The four versions were constructed according to the degrees of overlap of the dimensions, either partially or totally overlapping. This manipulation allowed us to observe to what extent the dimensions’ overlap might affect the participants’ performance. It is expected that the bidialectals will have longer response latencies than the monolinguals on verbal tasks, whereas when the verbal tasks involve a decrease in the negative effect of the lexical access and retrieval to a certain degree and depend heavily on greater inhibitory control and attentional control, bidialectals will tend to have shorter response latencies than the monolinguals.

Participants

The participants were identical to those in Experiment 1.

Apparatus

The apparatus and software used in this production experiment were the same as those in Experiment 1. The software E-prime controlled the presentation of stimuli, and using the response box, once the participants responded to the current trial (i.e., when they said aloud the test item), the next trial was automatically presented. The participants’ verbal response times were measured through the response box. The RT was determined by the elapsed time span starting from the showing of stimulus and ending at the onset of the saying of the word or the naming of the picture.

Stimuli and procedure

Before the experiment started, the participants were given an informed consent according to the guidelines of the human research ethics of National Cheng Kung University in Taiwan, as was done in Experiment 1. They were tested individually in a soundproof booth in the same visit as Experiment 1 and were paid \$25 for participating in this experiment. The testing of each participant in this experiment lasted about one and a half hours, with three 10-minute breaks. Before conducting the experiment, all of the bidialectals were instructed to use Mandarin, their less-dominant dialect, to perform all of the verbal tasks.

Experiment 2 consisted of four verbal variants of Stroop tasks, Number Stroop, Stroop Color and Word, Stroop day-night, and Stroop picture-naming. These verbal tasks involve different cognitive processes and loads, leading to different degrees of difficulty. They are ranked from low to high as Number Stroop (low), Stroop Color and Word (intermediate), Stroop day-night (high), and Stroop picture-naming (high) based on the dimensional overlapping degree (Kornblum et al., 1999) and attentional focus (Chen, 2003; LaBerge et al., 1991). The Number Stroop task comprises two partially overlapping dimensions and demands less capacity of cognitive control in comparison to the Stroop Color and Word task, such that the difficulty level of the former is lower than that of the latter. The Stroop Color and Word task is easier than the Stroop day-night and Stroop picture-naming tasks. Stroop Color and Word (where participants only remember to name the color-word’s ink color) requires less working memory and involves less lexical access²⁶ as compared with the Stroop day-night and Stroop picture-naming tasks (where participants need to memorize two task rules, “when seeing blue” and “when seeing red”). Moreover, Stroop day-night and Stroop picture-naming involve similar cognitive processes and loads and thus their difficulty levels are considered similar. Accordingly, the difficulty levels of Stroop day-night and Stroop picture-naming are ranked the highest, followed by Stroop Color and Word, and Number Stroop is the lowest.

²⁵ Before conducting the present investigation, we carried out two small-scale pilot studies, in each of which 8 participants (4 monolinguals and 4 bidialectals) took part, to prevent a floor effect and a ceiling effect from occurring. Through the two pilot studies, we modified some experimental designs and ruled out three tasks, two nonverbal tasks and one verbal task. Furthermore, the bidialectals seemed to perform on certain verbal tasks better than monolinguals did. However, due to a lack of big data, it was difficult for us to run statistical analysis.

²⁶ That is, the Stroop Color and Word task involves less difficulty in accessing lexical (from which bilingual disadvantage results).

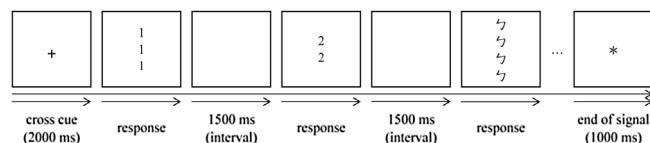


Fig. 3. Presentation of the stimuli for the Number Stroop task.

The four verbal Stroop tasks are depicted in (a) through (d) below. The order of the presentation of the four tasks was counterbalanced across participants.

- (a) *Number Stroop task*. This task was adapted from Rogers and Monsell (1995) and included 50 trials, which were presented in random order. In 20 out of these 50 trials, the digit stimuli and the number of the digit stimuli items were congruent (e.g., two 2s or four 4s are presented), in 20 trials, they were incongruent (e.g., three 1s or two 4s are presented), and in the remaining 10 trials, the stimuli were presented in Mandarin phonetic symbols (e.g., the symbol “ㄅ” corresponding to the phonetic alphabet [p]).²⁷ Mandarin phonetic symbols, not related to any number concept, were a control or neutral condition, with no concern to congruence or incongruence. In this task, the participants were asked to say aloud in Mandarin. In this task, the event presentation was as follows: (1) a fixation point (“+”) was shown on the center of the screen and remained for 2000 ms; (2) 10 practice trials were presented one after the other; (3) after the 10 practice trials, the experimental stimuli were presented, each experimental stimulus remained on the screen until the participant said the number of digits or phonetic notations, and the next trial was automatically started; (4) each test trial was followed by a blank screen with a 1500-ms onset-to-onset interval before the next trial started; and (5) after the presentation of the last stimulus, an asterisk “*” was shown for 1000 ms, which signaled the end of the block. Fig. 3 below presents the stimuli sequence in the numerical task:
- (b) *Verbal Stroop color and word task*. The stimuli in this task were color-words, namely purple, green, yellow, black, blue, and red. This task contained 40 trials, half of which were congruent and half incongruent. In each trial, one color-word was presented. In some trials, the name of the color-word and its ink color were congruent, such as the stimulus word “紫 (purple)” in Fig. 4 below presented in purple ink; in other trials, the color-word name and its ink color were incongruent, for example, the color-word “紅 (red)” in black ink. The participants were required to say aloud the ink color of the color-words in Mandarin. The presentation sequence of stimuli for this task was identical to that in the Number Stroop task, as shown in Fig. 4 below:
- (c) *Stroop day-night task (picture-naming)*. Gerstadt et al.’s (1994) research was adapted to design this task. In that study, because the participants were children, they considered the children’s ability and adopted only two conditions: night-day and cat-dog. However, in the current investigation, the participants were adults, and hence we raised the level of difficulty by including five conditions—sun-moon, cat-dog, fire-water, table-chair, and elder-child—and two kinds of trials—congruent and incongruent. In the congruent trials, the participants had to say aloud the name of the picture in Mandarin; for example, when shown the moon picture, they had to say “moon.” In the incongruent trials, the participants had to say aloud the opposite name of the picture; for example, when shown the moon picture, they had to say “sun” and vice versa. This task consisted of 10 pictures, each of which was displayed in a trial and framed by either a blue or red circle ($10 \times 2 = 20$). Each picture embedded in a blue or red circle was repeated twice ($20 \times 2 = 40$), and therefore there were a total of 40 trials. Before proceeding with the experiment, the participants were instructed to familiarize themselves with the target pictures and their Mandarin names; after that, they were given the rules for naming the pictures. The presentation of the stimuli in this task was the same as that in the Number Stroop task, as shown in Fig. 5 below:
- (d) *Stroop picture-naming task (saying aloud object names or “no”)*. This task was designed based on Costa and colleagues’ (2006, 2008) bilingualism studies, which utilized a picture-naming task. In their research, a picture was presented in either red or blue in each trial. When shown either of the two colors, English-German bilingual speakers were told which of their two languages to use to name the picture. For instance, when a picture in blue appeared, they had to say it in English, and when shown a red picture, they had to say it in German. In an attempt to compare the performance of the two language groups (i.e., monolinguals and bidialectals) in the current study, Costa and colleagues’ (2006, 2008) experiments were modified. In this task, all of the participants said the names of the pictures in Mandarin. The task consisted of 10 pictures and a total of 40 trials. Each picture was presented in either blue or red in each trial ($10 \times 2 = 20$), and each picture in either of the two colors was repeated twice ($20 \times 2 = 40$). Before the

²⁷ The phonetic symbols of Mandarin Chinese such as “ㄅ”, “ㄆ”, “ㄇ”, and “ㄏ” are also known as Zhuyin Fuhao or Bopomofo symbols utilized in Taiwan and visually represent speech sounds, hence they do not denote numerical numbers.

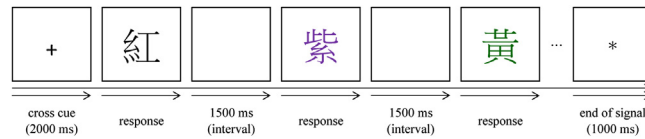


Fig. 4. Presentation of the stimuli for the verbal Stroop Color and Word task.

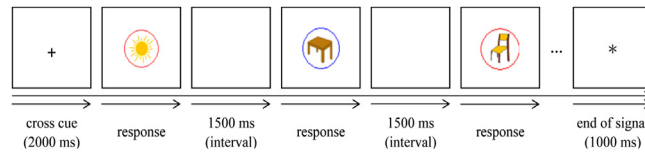


Fig. 5. Presentation of the stimuli for the Stroop day-night task.

experiment, the participants were asked to familiarize themselves with all of the target pictures and their Chinese names; after that, they were given the rules for picture-naming: when the picture was presented in blue, they had to say aloud its Mandarin name; if the picture was red, they should not say it but instead say *meiyou* “沒有 (no)” in Mandarin. This change would require “more” inhibitory control when compared with the demand for inhibitory control in Costa et al's bilingual study. The bidialectal participants had to inhibit not only the words that might have been activated in the target and non-target dialects to name the picture but also the interference from “*bo* (no)” in Minnan or “*mo* (no)” in Hakka (their dominant dialect), the corresponding words of the response “*meiyou* (no)” in Mandarin.

In this task, the trial presentation proceeded as follows: (1) on the screen, a solid blue circle and a solid red circle appeared successively, along with the words *shuo zhongwen* “說中文 (to name the picture in Mandarin)” and *shuo meiyou* “說沒有 (to say ‘no’ in Mandarin)”, respectively, and a circle was presented for 3000 ms; (2) 10 practice trials appeared in a row; (3) the picture disappeared after the participant named the picture and the next trial started automatically; (4) after the 10 practice trials, the experimental pictures were presented on the screen until the participant responded verbally; (5) each target picture is followed by a blank screen with a 1500 ms onset-to-onset interval, and after the blank screen, the next test picture appeared and so on; and (6) after the last picture was presented, an asterisk “*” appeared for 1000 ms, indicating the end of the task. Fig. 6 below shows an example of the presentation sequence:

Measurement of response latencies

The measurement of the response times in this experiment was the same as that in Experiment 1, except that the participants' verbal RTs were recorded via the voice key installed on the response box connected to the computer. Only correct responses were measured and errors were excluded from the measurement of response latencies. For example, in the Stroop picture-naming task, the participants mistakenly said “no” when the blue circle was presented and the wrong responses were ruled out from the analysis.

Recording and measurement

There were two personal computers utilized in Experiment 2. One computer was used for presenting stimuli and the other for recording the participants' speech. The participants were required to wear Shure SM10A headset connected to the computer. Their speech was directly recorded into the computer using Speech Analyzer 3.0.1 software. In the meantime, a professional digital pen recorder (Olympus DS-20) was used for backup. Each recorded item was listened to and then wrong responses were recognized and removed from the measurement of response latencies.

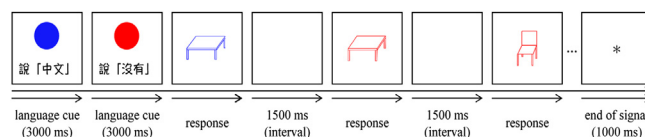


Fig. 6. Presentation of the stimuli for the Stroop picture-naming task.

Table 7

The two-way mixed ANOVA table for Experiment 2.

	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i>	η^2
Language Group (independent factor)	1	59598.40	2.580	.117	.064
Tasks (dependent factor)	3	2044954.20	169.157	.000*	.817
Language Group $\times$ Tasks	3	133659.40	11.056	.000*	.225

Results of Experiment 2

To examine whether there was a statistically significant difference in RTs between the two language groups across all four verbal tasks, a two-way 2 (language groups: monolingual vs. bidialectal, between subjects) $\times$ 4 (different tasks, within subjects repeated measures) mixed ANOVA was conducted. The four tasks consisted of the Number Stroop task, the verbal Stroop Color and Word task, the Stroop day-night task, and the Stroop picture-naming task (hereafter Number Stroop, Stroop CW(v), Stroop DN, and Stroop PN). The monolinguals and bidialectals did not differ in response error rates significantly ($p > .05$), suggesting no speed-accuracy tradeoff in the tasks.

A two-way ANOVA on these response latencies for language group and task indicated a significant interaction of the two factors, $F(3,114) = 11.056$, $MSE = 133659.40$, $p < .001$, $\eta^2 = .225$ (see Table 7). The interaction indicated that the language background effects might differ in the different tasks.

The follow-up Bonferroni tests on language background simple effect showed that when performing Stroop CW(v) (T2) and Stroop DN (T3), there were significant differences in RTs between language groups, where RTs of the bidialectals were shorter than those of the monolinguals (T2: bidialectal $M = 857.50$ ms, monolingual $M = 1047.00$ ms, $p < .001$; T3: bidialectal $M = 948.65$ ms, monolingual $M = 1005.65$ ms, $p = .034$), indicating the presence of a bidialectal advantage. The task simple-effect test indicated that for either of the two language groups, there were significant differences in RTs between different difficulty levels of the tasks (Table 8).

Moreover, the pairwise comparisons on the task simple effect showed that for the monolinguals, there were significant differences in RTs between different tasks. The RTs for the Number Stroop task (T1) were significantly faster than those for each of the other three tasks (T1 vs. T2: $p < .001$; T1 vs. T3: $p < .001$; T1 vs. T4: $p < .001$) and those for Stroop DN (T3) were significantly faster than Stroop PN (T4) (T3 vs. T4: $p = .034$), with no significant differences in the RTs between T2 and T3 ($p = .981$) and between T2 and T4 ($p = .684$). For the bidialectals, there were significant differences between RTs for all four verbal tasks (all $p < .05$), with RTs for T1 faster than T2, which in turn was faster than T3, which in turn was faster than T4 (T1 vs. T2: $p < .001$; T1 vs. T3: $p < .001$; T1 vs. T4: $p < .001$; T2 vs. T3: $p = .020$; T2 vs. T4: $p < .001$; T3 vs. T4: $p < .001$). These statistical results indicated that for either of the two language groups, their response latencies between the different difficulty levels of the tasks were different over all, along the same ranking order of difficulty levels as predicted. Table 9 below shows the means of RTs of the two language groups across the four verbal tasks.

Discussion of Experiment 2

Experiment 2 was conducted to explore whether the bidialectal advantage as shown in the nonverbal task with the highest difficulty level in Experiment 1 might also be observed in the verbal tasks, which were hypothesized to be more difficult than many nonverbal tasks in general. The results obtained in this experiment did not completely replicate the pattern of findings observed in Experiment 1. There was a statistically significant interaction between task and language group, implying that there were no general language effects across all tasks, namely the bidialectal advantage

Table 8

Simple effects and pairwise comparison for Experiment 2.

Source	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i>	Pairwise comparisons
<i>Language Group (independent factor)</i>					
T1-Number Stroop	1	921.60	.145	.705	
T2-Stroop CW(v)	1	359102.50	17.126	.000*	Monolingual > Bidialectal
T3-Stroop DN	1	32490.00	4.854	.034*	Monolingual > Bidialectal
T4-Stroop PN	1	68062.50	2.685	.110	
<i>Tasks (dependent factor)</i>					
Monolingual	3	1031961.833	113.462	.000*	T2, T3, T4 > T1; T4 > T3 > T1
Bidialectal	3	1146651.767	76.023	.000*	T4 > T3 > T2 > T1

Note: T1 = Number Stroop task; T2 = Stroop CW(v) task; T3 = Stroop DN task; T4 = Stroop PN task; ">" = longer RTs than; * $p < .05$.

Table 9

Mean response times of the monolingual and bidialectal participants in the four verbal tasks.

	Difficulty level	Monolingual (N = 20)		Bidialectal (N = 20)		All participants (N = 40)	
		Mean (ms)	SD	Mean (ms)	SD	Mean (ms)	SD
T1-Number Stroop	Low	612.70	72.14	622.30	86.63	617.50	12.60
T2-Stroop CW(v)	Intermediate	1047.00	177.60	857.50	101.96	952.25	22.90
T3-Stroop DN	High	1005.65	82.44	948.65	81.19	977.15	12.94
T4-Stroop PN	High	1119.05	89.10	1201.55	206.79	1160.30	25.18

Note: N = number of participants; ms = millisecond, the unit for response latencies; SD = Standard Deviation.

(language background effect) did not exist across the board. Follow-up simple-effect tests showed that the bidialectal advantage existed in the two middle difficulty level tasks, Stroop CW(v) (T2) and Stroop DN (T3) and there were no bidialectal advantages in Number Stroop (T1), the easiest of the four tasks, and Stroop PN (T4), the most difficult of the four tasks. One potential explanation for this similarity is that T1 reaches the floor, and T4 reaches the ceiling, and therefore no language group differences can be shown.

The similarity between the monolinguals and bidialectals in Number Stroop (T1) and Stroop PN (T4) suggests that dialectal advantages show up only in tasks with appropriate task difficulties.²⁸ For T1, its processing demand for cognitive control might not be challenging enough for participants, and therefore the subtle cognitive advantage of the bidialectals cannot be shown. Conversely, T4 is too difficult for both language groups to complete. Thus, the significant difference between the two groups (i.e., the bidialectal advantage) was not found in these two tasks. It was found that the bidialectals performed better than the monolinguals on the two verbal tasks, Stroop CW(v) (T2) and Stroop DN (T3). Past studies generally found bilingual disadvantages in the verbal tasks, but the result of T2 and T3 showed the opposite, and thus this was a finding of theoretical interest. The bidialectal advantage revealed that the bidialectals were more efficient in inhibiting interference from the distractors in these two verbal tasks relative to the monolinguals.²⁹

Although Stroop DN (T3) and Stroop PN (T4) had the same high difficulty level based on previous studies (e.g., Gerstadt et al., 1994; Levell et al., 1999),³⁰ the results of the present study showed that T2 and T3 had the same difficulty level (given that the response times can be an indicator of task difficulty levels), and T4 was significantly higher in difficulty level than either T2 and T3.³¹ Thus the finding obtained in the current study challenges the claims of the past studies.

The reason for why Stroop PN (T4) is more difficult is that the task design requires not only the desired inhibitory control, but also unfairly requires the bidialectals (but not the monolinguals) to not use the words of their own dominant dialects in their verbal responses. It is probably a defect in methodological design.³² Through a closer inspection of the participants' feedback during and after the experiments and the errors they made, the similarity between the bidialectals and monolinguals in T4 might be explained from the perspective of limited attentional resources: increased attention for effortfully remembering the task rules depletes resources for inhibitory control. Further examination of participants'

²⁸ As Costa and colleagues (2009) pointed out, different task demands lead to the fact that the bilingual advantage in cognitive control could not be identified in certain conditions; that is, under different conditions of a task, sometimes a bilingual advantage can be observed, but at times it cannot.

²⁹ As assumed in the previous section "Discussion of Experiment 1," given that the interference from lexical access (which causes the bidialectal disadvantage) has been reduced from verbal tasks to a certain degree, it is expected that a bidialectal advantage is more likely to be identified. As expected, the bidialectal advantages were found in Stroop DN (T2) and verbal Stroop CW (T3), but not in Number Stroop (T1) and Stroop PN (T4). In terms of the task difficulty levels, T1 and T4 indicated the floor and ceiling effects, respectively, leading to the equivalent performance between monolinguals and bidialectals.

³⁰ To be precise, according to the earlier research, it was not easy to explicitly distinguish the comparative difficulty levels between Stroop DN and Stroop PN. As a result, based on their major characteristics, the two tasks might be considered those with similar difficulty levels.

³¹ To our surprise, Stroop PN was the most difficult of the four verbal tasks for all of the participants, as reflected in the response latencies. The result was just unexpected, but this was why the results of this study were interesting and made a contribution to the fields such as bilingualism and cognitive control: the factor of the "response item" (saying "NO" or others) should be taken into account when researchers design experiments in future studies.

³² In future studies, we can examine what difference there is in response times between monolinguals and bidialectals when the bidialectals' dominant dialect is used as a verbal response in Stroop picture-naming tasks. However, it should be noted that it was not a mistake in the method design to require the bidialectals to use their NON-primary dialect "Mandarin" in responses. As pointed out in the section of Method, in response to the stimuli (i.e., Chinese characters which correspond to Mandarin sound system), only the dialect, Mandarin, can be chosen to use in this study so as to compare the performance differences between monolinguals and bidialectals. On the other hand, in so doing, a potential translation factor could be excluded from the experiments.

feedback reports shows that during their performance of practice trials, at times, the Minnan-Mandarin bidialectal adults mistakenly uttered the bisyllabic “*meiyou* (no)” in Mandarin (the target dialect) as a monosyllabic “*bo* (no)” in Minnan (the non-target dialect), and the Hakka-Mandarin bidialectal adults mispronounced it as “*mo* (no)” in Hakka (the non-target dialect). Consequently, the bidialectal participants had been offered more practice trials until they did not make these mistakes, and in so doing, no errors of “*bo* (no)” or “*mo* (no)” were discovered when performing the experimental trials. In addition, the bidialectals reflected that it was difficult for them to say “*meiyou* (no)” instead of “*bo* (no)” or “*mo* (no)”. They not only had to inhibit interference from the equivalent words “*bo* (no)” and “*mo* (no)” in their dominant dialect (i.e., non-target dialect) but also had to remember to say the target word “*meiyou* (no)” when not naming the picture.

Moreover, in Stroop PN (T4), working memory seems to have high priority over inhibitory control. Such a condition is unfavorable to the bidialectals because they are hypothesized to have enhanced inhibitory control relative to the monolinguals. The implicit evidence comes from the participants’ feedback reports. Half of the participants pointed out that saying “*meiyou* (no)” was more difficult than saying the opposite name of the picture as required in Stroop DN (T3) as they often “almost forgot” to say aloud “no” when seeing the pictures in red. Interestingly, out of 40 participants, eight clearly reported that it was considerably difficult to say “no” because they “did see” the pictures presented on the screen; therefore, they had to restrain their impulses to name the current picture and “remember” to say “no” instead.³³ In this view, participants tended to prioritize more attentional resources for memorizing the task rules in T4, causing fewer resources left for inhibitory control, as the attentional resources are limited. When performing T4, the participants tended to consume more attentional resources for remembering the task rules to ensure that they would not forget to respond to the current stimulus and make a correct response as much as possible. Such increased attention for the memory of the task rules in T4 certainly reduced resources left for inhibitory control.

Comparing Stroop CW(v) (T2), DN (T3), and PN (T4) with the observations mentioned above, T4 involves greater memory load than T2 and T3 do. Specifically, besides the demand for inhibitory control,³⁴ T4 where the bidialectals used their weaker dialect in answers rests more on “working memory” than T2 and T3 do.³⁵ In terms of limited attentional resources, ample resources can be utilized to operate inhibitory control in both T2 and T3. Conversely, in T4, the resources fail to smoothly manage inhibitory control under the high memory load condition. In T4, given that most of the resources have been allocated to memorize the task rules and maintain them, then only a few resources will be left for the operation of inhibitory control; in other words, if the conscious remembering of the saying “*meiyou* (no)” consumes many resources, fewer resources will remain for the subsequent inhibitory process. This might be a possible cause resulting in the differences, namely why the bidialectal advantages were present only in T2 and T3, but absent in T4. However, whether this interpretation is viable still awaits future investigation.

General discussion

The current investigation aimed to examine whether past findings of bilingual advantages and disadvantages identified in bilinguals could be unfolded in the performance of an older bidialectal population. In addition, this study attempted to explore to what extent and which conditions of tasks “bidialectal advantages” could be generalized. To answer these questions, four nonverbal and four verbal cognitive tasks were conducted, and the differences in response latencies were compared between the older monolingual and bidialectal adults.³⁶ To facilitate a clear understanding of the overall larger picture, a summary of findings across all tasks is provided in Table 10 below.

³³ More interestingly, one Minnan-Mandarin bidialectal expressed that she did not like to say “no (*meiyou*沒有, *busing*不行, and *buneng*不能)” in her own communication mode, but in order to perform the Stroop PN task successfully, she “tried very hard to remember” to say “*meiyou* (no)” without forgetting to make a correct response to the picture in red. Additionally, one Hakka-Mandarin bidialectal suggested that the color “red” of the stimuli should be replaced with other colors like “green” because for him, the color “red” connoted “STOP” as what a red signal light meant, which deterred him from responding to the red stimuli, leading to the fact that he often had to “remember” to make a response. All of these participants’ opinions can contribute to the experimental design in future studies.

³⁴ In T4, the participants had to suppress the words that might have been activated in the target and non-target dialects to name the picture.

³⁵ According to the participants’ feedback reports and the errors they made, it was reported that in T2 and T3, “inhibiting” the prepotent but non-target response was more difficult than “remembering” the task rules. Answering the question of why they had made the errors, the participants reported that they “remembered” the rules but failed to inhibit the irrelevant prepotent response. Therefore, among different executive functions, T2 and T3 (both of which showed bidialectal advantage) might tend to be more concerned with inhibitory control as compared to T4; the more the tasks involve more inhibitory control, the better the bidialectals may perform, as speculated from the results of Experiment 1. However, this speculation that the difficulty may be related to the different demands for cognitive abilities will be the issue for future studies.

³⁶ The two types of tasks have been designed to examine how differently monolinguals and bidialectals performed in nonverbal and verbal tasks. According to the previous literature (e.g., Bialystok et al., 2008; Blom et al., 2014; Gollan et al., 2007), a bidialectal advantage is expected to be present in the nonverbal tasks, whereas a disadvantage is predicted to emerge in the verbal tasks. We further manipulated the lexical factor to examine whether the bidialectal advantage would be more likely to emerge in the verbal tasks; that is, we ruled out the factor of lexical access and retrieval (one of the main causes resulting in the disadvantage) to observe if the bidialectal advantage would show up in the verbal tasks.

Table 10
A summary of results across all tasks.

Task		Difficulty level	Monolingual RT means (ms)	Bidialectal RT means (ms)	Monolingual-bidialectal difference (ms)	Difference <i>P</i> value
Nonverbal	Flanker	Low	587.45	582.95	4.5	.905
	Simon CS	Low+	634.00	595.55	38.45	.248
	1-back	Intermediate	700.80	692.20	8.6	.777
	Stroop CW	High	1115.25	1033.45	81.8	.047*
Verbal	Number Stroop	Low	612.70	622.30	-9.6	.705
	Stroop CW	Intermediate	1047.00	857.50	189.5	.000*
	Stroop DN	High	1005.65	948.65	57	.034*
	Stroop PN	High	1119.05	1201.55	-82.5	.110

Note: **p* < .05.

Effects of the tasks

Task difficulty levels have been hypothesized as a possible explanation for the presence of bilingual advantage in some cognitive tasks and absence of such advantage in others. Previous literature has reported that individuals may perform differently on similar tasks varying in task difficulty (Costa et al., 2009; Dong and Li, 2015). Dong and Liu (2016), based on their findings in Number Stroop and n-back tasks, claimed that the degree of processing demand played an important role in determining whether a bilingual advantage unfolded or not. Van den Noort and colleagues (2019) further pinpointed that it was difficult to identify bilingual advantages in young adults who were at the peak of their cognitive capacities in Simon tasks because the Simon task is one of the easiest tasks, resulting in the floor effect. These past studies provide evidence that bilingual advantage can only be unveiled in tasks with appropriate levels of cognitive demands, not too easy or too difficult. Consider Table 11 displaying a summary of conditions for the presence of bidialectal advantage identified in the present study.

In the current research, there are three levels of task difficulties among tasks. Among the nonverbal tasks in Experiment 1, two are easy (Simon CS and Flanker), one intermediate (1-back), and one difficult (Stroop CW(nonv)). Among the verbal tasks in Experiment 2, one is easy (Number Stroop), one intermediate (Stroop CW(v)), and two difficult (Stroop DN and Stroop PN). The different task difficulty levels are hypothesized to have resulted from the major executive functions, including inhibitory control, attentional control, and working memory, involved in these tasks.

Bidialectal advantages were found in one nonverbal task (Stroop CW(nonv)) and two verbal tasks (Stroop CW(v) and Stroop DN). These three tasks were rated as either intermediate or high. No bidialectal advantages emerged in the remaining tasks. Among these tasks, the difficulty levels, except for Stroop PN (high), were either low (Flanker, Simon CS, and Number Stroop) or intermediate (1-back). The Stroop PN task did not show bidialectal advantage even though its difficulty level was rated as high. A closer examination of the participants' feedback showed that Stroop PN required much more memory demand than we had expected, in addition to the intended inhibitory controls, as compared to Stroop CW (nonv), CW(v), and DN. This unintended excessive memory demand might be one possible account for why bidialectal advantage was present in Stroop CW(nonv), CW(v), and DN, but not in Stroop PN. In other words, the bidialectal advantage observed in high difficulty level tasks may not be shown if the task difficulty level is excessively high. Taken together, these findings revealed that the presence of bidialectal advantage was more sensitive to the condition with a higher level of task difficulty, as long as the difficulty level is not excessively high.

Table 11
A summary of conditions for the presence of bidialectal advantage.

Task		Difficulty level	Presence of advantage
Nonverbal	Flanker	Low	✓
	Simon CS	Low+	
	1-back	Intermediate	
	Stroop CW(nonv)	High	
Verbal	Number Stroop	Low	✓
	Stroop CW(v)	Intermediate	
	Stroop DN	High	
	Stroop PN	High	

Effects of bidialectalism

Mixed but interesting results were found in the present research. Among the four nonverbal tasks, a bidialectal advantage was uncovered only in Stroop CW(nonv), but the other three nonverbal tasks showed no advantage. These findings do not support Hypothesis 1 but support Hypothesis 2: bidialectal advantages are more likely to be found in more demanding (i.e., more difficult) nonverbal tasks. Bidialectals performed significantly better than monolinguals in the most difficult task, Stroop CW(nonv), but such bidialectal advantage was not found in the other three nonverbal tasks—Flanker, Simon CS, and 1-back, which were either easy or intermediate in difficulty levels.

This finding does not lend support to the studies which found bilingual advantages in such tasks. Costa et al. (2009) discovered a bilingual advantage in young adults on a flanker task with a “high-monitoring” condition; Bialystok (2006) reported that young bilingual adults performed better than monolinguals on a Simon task; and Teubner-Rhodes et al. (2016) found that young bilingual adults exceeded their monolingual counterparts on a high-conflict version of the n-back task (but not on a no-conflict version). However, the findings of the three nonverbal tasks in the current study are in line with one of the findings in Costa et al.’s (2009) study in which bilinguals and monolinguals had similar performance in the “low-monitoring” condition. Further review of the difference and similarity between this study and the previous research shows that the difficulty levels of the three tasks utilized in the current study are either low or intermediate (see Table 1); in a certain sense, the conditions manipulated in the three task are similar to the low-monitoring condition in Costa et al.’s study. Accordingly, the difficulty level of tasks might be the reason for the similarity and difference between this study and the earlier research. In other words, bilingual or bidialectal advantages only appear in tasks with adequate cognitive challenges, and do not appear in tasks with low levels of difficulty.

In contrast to the three nonverbal tasks discussed above, Stroop CW(nonv) showed bidialectal advantage. This finding challenges the finding of Kousaie and Phillips (2012), where no such advantage was found in the nonverbal Stroop CW task among the “older” bilingual adults (mean age: 71.9 years). Two possible explanations are provided below for the conflicting findings between this study and Kousaie and Phillips’ (2012) study.

The first possible reason is related to the relative strengths of the links between semantics and phonology for the target lexical items in the two studies. In this study, the bidialectals’ mapping between semantics and phonology for the target words is weaker because the target lexical items were in their non-dominant dialect, Mandarin, which allowed them to easily escape from the influence of the predominant and automatic word-reading (non-target) and pay much attention to the color-word’s ink (target). Consequently, the bidialectals needed less inhibitory control in performing the task. In contrast, the semantics-phonology mapping for the monolinguals is stronger than that for the bidialectals, as the target lexical items were in their dominant dialect (which is their only dialect). As a result, the monolinguals had to be confronted with higher inhibitory control challenges than the bidialectals in this study.³⁷ In other words, due to the design of this study, the task difficulty levels were not equal for the two language groups. In Kousaie and Phillips’ (2012) study, in contrast, the stimulus language, English, was the dominant language for both the monolinguals and the bilinguals, and therefore the two language groups in that study were confronted with tasks of the same difficulty levels. Hence, the weaker link between semantics and phonology due to the use of the non-dominant dialect as the stimulus for the bidialectals in this study might be a possible reason for why bidialectals outperformed monolinguals in the nonverbal Stroop CW task.

It should be noted that if the lexeme-phoneme mapping explanation is corroborated, it implies that this explication defeats the original claim about that bidialectals have better cognitive control abilities than monolinguals. This mapping account suggests that it is the unfair design of the Stroop CW(nonv) task, which presents a task that is easier for bidialectals than for monolinguals (because the latter need to inhibit the stronger mapping), rather than bidialectals’ enhanced cognitive control, that facilitated the bidialectals to outperform the monolinguals in this study. To rule out this possible cause, future studies are required to ensure the equal task difficulty levels for the two language groups in the design.

However, even though it makes intuitive sense that the use of the non-dominant dialect/language may present less challenge for inhibitory control due to the weaker lexeme-phoneme mapping connection, many past studies reported that bilinguals’ two languages are always co-activated even in a unilingual scenario³⁸ (i.e., even when only one language is used in a conversation, bilinguals’ two languages are still co-activated), causing inevitable interference from the co-activation of linguistic representations of two languages (Bialystok, 2007; Bialystok and Craik, 2010; Blumenfeld and Marian, 2007; Costa and Santesteban, 2004; Libben and Titone, 2009; Marian et al., 2003; Shook and Marian, 2013). According to the co-activation perspective, the bidialectals in the current research would still need to suppress

³⁷ According to Green’s inhibitory control model (1998), the baseline level of activation of Mandarin lexical representations is higher for the monolinguals than for bidialectals, leading to the fact that the monolinguals must inhibit Mandarin lexical items (non-target) more strongly in order to name the color of the word’s ink (target) successfully.

³⁸ A unilingual scenario refers to a context where only one language is needed for communication.

interference from their dominant D1, either Minnan or Hakka. Thus, it is reasonable and valid to assume that the unbalanced bidialectals in this study needed the same amount of inhibition to suppress irrelevant information as did the monolinguals, since their dominant dialect (either Minnan or Hakka) is co-activated during the task.

Let us now turn to the findings of the verbal tasks. In the four verbal tasks, this study found that bidialectals performed significantly better than monolinguals in the two tasks, Stroop CW and Stroop DN, and performed as well as the monolinguals in the two other tasks, Number Stroop and Stroop PN. These results showed a clear contrast to previous studies (Gollan et al., 2007; Oller and Eilers, 2002; Ransdell and Fischler, 1989) that found bilinguals' performance was significantly more poorly than that of the monolinguals. These findings rejected both Hypothesis 3 "the bidialectal disadvantage will be found in the verbal tasks" and Hypothesis 4 "the more heavily a verbal task relies on inhibitory control and attentional control, the more likely the bidialectal advantage will show up." According to the results of this study, among the four verbal tasks, Stroop PN was the most difficult task, followed by Stroop DN and Stroop CW(v), and Number Stroop was the easiest. The easiest and the most difficult tasks showed no statistically significant difference, but the two middle ones showed the difference, suggesting that the effects (cognitive advantages) depend on the right level of task difficulty. That is, no bidialectal advantage was identified in too low or too high difficulty levels, whereas the advantage was discovered in middle level difficulties. These results did not support Hypotheses 3 and 4.

Significant bidialectal advantages were observed in the verbal Stroop CW and Stroop DN tasks. This finding of bidialectal advantages in the verbal tasks conflicts with findings in earlier research in which bilingual disadvantages were commonly observed in verbal tasks. In Rosselli and colleagues' (2002) three verbal Stroop conditions (i.e., color-reading, color-naming, and color-word conditions), a bilingual disadvantage was found in the color-naming condition,³⁹ and in the other two tests, bilinguals and monolinguals performed equivalently.⁴⁰ The findings of the current study challenge the findings of their study. The finding that bidialectal cognitive advantages can be observed only in certain conditions (i.e., in tasks with reasonably high cognitive difficulties), but not others (i.e., if the task is too easy or excessively difficult), shows that future research on bilingual and bidialectal cognitive advantages should consider the task difficulty level, and specially pinpoint what kind of cognitive challenges and what level of such challenges would be.

Conclusion and implications

The present study adopted four nonverbal and four verbal cognitive tasks with different task difficulty levels to explore whether the cognitive advantages among the bilinguals were also found among bidialectal elder adults, and whether such cognitive advantages were consistently found across verbal and nonverbal tasks and different task difficulty levels. The results indicated that bidialectal advantages were identified in the older Minnan-Mandarin and Hakka-Mandarin bidialectals on one nonverbal test, Stroop CW(nonv), and two verbal tests, Stroop CW(v) and Stroop DN, which had adequate levels of cognitive difficulty, but not the other two verbal tasks which were either very easy or excessively difficult. These findings provide three contributions to the study of the relationships between bilingualism/bidialectalism and the speakers' executive functioning. First, these results show that the advantages in cognitive control extended beyond bilingualism to bidialectalism, and that such cognitive advantage found in young adult population were also maintained in the elder adult population. Additionally, the results of the study show that such cognitive advantages were found in both nonverbal and verbal tasks, challenging the findings of past studies. Lastly, the results show that the bidialectal cognitive advantages were only found in tasks with certain task difficulty levels, when the tasks were challenging enough, but not too easy or too difficult.

The findings of this study also point at some new directions for future research. Cognitive tasks should be more thoughtfully designed to ensure equal task difficulty levels for all the comparison language groups. Different cognitive tasks can be more carefully selected so as to tease out which specific cognitive skills (e.g., inhibition, attention, or memory span) receive benefit from bidialectalism. In addition, the same tests and procedures can be applied to strategically planned age groups to examine more accurately the developmental effects of bidialectal advantage effects.

³⁹ Rosselli and colleagues (2002) classified their bilingual participants into four subgroups: high-proficient balanced bilinguals, low-proficient balanced bilinguals, English dominant unbalanced bilinguals, and Spanish dominant unbalanced bilinguals. Along the lines of Rosselli and colleagues' (2002) research methods, the methods utilized to collect the data in the present study were almost identical: "unbalanced" bidialectals were recruited, the verbal Stroop Color and Word Test was adopted, and the stimuli was presented in the bidialectals' weaker dialect, Mandarin. However, in their study, Rosselli and colleagues did not compare the differences between monolinguals and "balanced" bilinguals or between monolinguals and "unbalanced" bilinguals, but just compared the differences between monolinguals and bilinguals as well as between balanced bilinguals and unbalanced bilinguals. Therefore, it is not clear what differences there are between monolinguals and "unbalanced" bilinguals in the three verbal Stroop conditions.

⁴⁰ In terms of the language effect, young unbalanced Spanish-English bilinguals were found to perform significantly slower in all three verbal Stroop conditions in their "less dominant" English than their dominant Spanish. Likewise, the English-dominant bilinguals were also slower in the Stroop tests in Spanish than in English.

Conflict of interest

The author declares that she and the institute supporting this work have no conflict of interest.

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Minnan-Mandarin and Hakka-Mandarin Bidialectal Participant Demographics

Table A1 below displays the demographic information of two bidialectal subgroups (D1 Minnan–D2 Mandarin and D1 Hakka–D2 Mandarin) with respect to male-to-female ratio, age range, mean age, the onset of D2 acquisition (D2 AOA), percent daily use of D2, D1 proficiency, D2 proficiency, and MMSE (the Mini-Mental Status Exam), with the standard deviation shown in parentheses.

Table A1
Elderly bidialectals' demographic and neurological characteristics.

	Bidialectal (N = 20)		Independent samples <i>t</i> -test		
	D1 Minnan-D2 Mandarin (N = 10)	D1 Hakka-D2 Mandarin (N = 10)	<i>df</i>	<i>t</i>	<i>p</i>
Male-to-female ratio	M7: F3	M6: F4			
Age range (in years)	61.00–74.00	62.00–75.00			
Mean age (in years)	68.00 (4.71)	68.70 (4.50)	18	–.340	.702
D2 AOA (in years)	11.10 (2.13)	11.90 (2.13)	18	–.839	.753
Daily use of D2 (percent)	39.5%	41.5%	18	–.682	.504
D1 proficiency	5.00 (.00)	5.00 (.00)			
D2 proficiency	Listening 4.10 (.32)	4.20 (.42)	18	–.600	.232
	Speaking 3.40 (.52)	3.60 (.52)	18	–.866	1.000
	Reading 3.20 (.42)	3.30 (.48)	18	–.493	.334
MMSE	26.30 (.95)	26.50(1.08)	18	–.440	.549

Note: N = number of participants; standard deviations in parentheses; D2 proficiency = participants' self-assessed language proficiency in three language skills (listening, speaking, and reading) in total points.

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